

figure 17-1
Client record card.

PERMANENT WAVE RECORD					
Name		Tel			
Address					
City		State		Zip	
DESCRIPTION OF HAIR					
Length ☐ short ☐ medium ☐ long	Texture ☐ coarse ☐ medium ☐ fine	Type ☐ normal ☐ resistant ☐ tinted ☐ highlighted ☐ bleached	Porosity ☐ very porous ☐ moderately porous ☐ normal	☐ slightly porous ☐ resistant	
CONDITION					
☐ very good ☐ good ☐ fair ☐ poor ☐ dry ☐ oily					
Tinted with					
Previously permed with					
TYPE OF PERM					
☐ alkaline ☐ acid ☐ body wave ☐ other					
No. of rods		Lotion		Strength	
RESULTS					
☐ good ☐ poor ☐ too tight ☐ too loose					
Date	Perm Used	Stylist	Date	Perm Used	Stylist
.....					
.....					
.....					
.....					
.....					

Any of these conditions can make a chemical process uncomfortable and dangerous to a client. Do not proceed with the service if abrasions or signs of scalp disease are present. Refer the client to a physician as necessary.

The hair analysis includes observing and determining the hair's:

- porosity
- texture
- elasticity
- density
- length
- direction of growth

HAIR POROSITY

How does porosity affect chemical texturizing services?

The processing time for chemical texturizing services depends more on hair porosity than on any other factor. Generally, the more porous the

hair, the less the processing time. The porosity level of the hair determines the speed with which moisture will be absorbed into the hair and is directly related to the condition of the cuticle layer. Hair porosity is classified as resistant, normal, or porous. These classifications help determine the most appropriate strength of a chemical product to use on different hair types.

- *Resistant* hair has a tight, compact cuticle layer that resists penetration of chemical solutions. This hair type requires a more alkaline solution to raise the cuticle and permit uniform saturation and processing.
- *Normal* porosity means that the hair is neither resistant nor overly porous. Chemical texture services performed on this hair type will usually process as expected.
- *Porous* hair has a raised cuticle layer that easily absorbs chemical solutions. This hair type requires a less alkaline solution to help prevent damage to the hair.

Do not proceed with the service if the hair shows signs of breakage or overporosity.



DID YOU KNOW?

Porosity is best evaluated on clean, dry hair. Roughness indicates a raised cuticle and a higher porosity level.

Porosity Level Mini-Procedure

To accurately test the porosity level:

1. Start by using these four different areas:

- at the front hairline
- temple
- near the crown
- nape

2. Grasp some strands of dry hair.

3. Hold the ends firmly with the thumb and index finger of one hand, and slide the fingers of the other hand from the ends toward the scalp.

4. If the fingers do not slide easily, or if the hair ruffles up as your fingers slide down the strands, the hair is porous. The more ruffles that form, the more porous the hair; the fewer ruffles that form, the less porous it is.

5. If the fingers slide easily and no ruffles are formed, the cuticle layers lie close to the hair shafts. This type of hair is the least porous and most resistant, and may require a longer processing time (Figure 17-2).

6. Keep in mind the porosity may be different from the base area to the ends of the hair. If drastically different, the client may need a product, such as a light spray moisturizer, to even out the porosity before proceeding with the service.

figure 17-2
Porosity test.



HAIR TEXTURE

How does hair texture affect chemical texturizing services?

- While hair porosity is important in determining the processing time, hair texture also plays a part in the decision. For example, coarse, porous hair will usually process faster than fine, resistant hair.
- Hair texture describes the diameter of a single strand of hair as being coarse, medium, or fine:
 - *Coarse* hair usually requires more processing than medium or fine hair and may also be more resistant to chemical processes.
 - *Medium* hair is the most common hair texture. It is considered normal and does not usually pose any special processing problems.
 - *Fine* hair is typically more fragile, easier to process, and more susceptible to damage from chemical services. Generally, fine hair will process faster and more easily than medium or coarse hair types.

HAIR ELASTICITY

How does hair elasticity affect chemical texturizing services?

- Hair elasticity is an important factor to consider because it is an indication of the strength of the cross-bonds in the hair.
- The greater the degree of elasticity, the longer the wave will remain in the hair because less relaxation of the hair occurs; thus the elasticity of the hair determines its ability to hold a curl.
- Hair elasticity is classified as normal or low:
 - *Normal* elasticity is indicated by wet hair that can stretch up to 50 percent of its original length and then return to that length without breaking. Hair with normal elasticity usually holds the curl from wet sets and permanent waves.
 - *Low* elasticity is indicated by wet hair that does not return to its original length when stretched and may not be able to hold curl patterns.

Test for Elasticity Mini-Procedure

To test for elasticity:

1. Take a few strands of damp hair.
2. Hold them between the thumb and forefinger of each hand.
3. Slowly stretch the strands between your fingers (**Figure 17-3**).
 - a. The more they can be stretched without breaking, the more elasticity the hair has.
 - b. If the elasticity is good, the hair slowly contracts after stretching.
 - c. Hair with poor elasticity will break quickly and easily when stretched.

figure 17-3
Elasticity test.



- d. Signs of poor elasticity include limpness, sponginess, and hair that tangles easily. Generally speaking, such hair will not develop a firm, strong wave and may result in breakage.
 - e. There are special waving solutions available for hair with poor elasticity; if used in combination with smaller-diameter rods, it may result in a satisfactory permanent wave or curl reformation process.
-

HAIR DENSITY

How does hair density affect chemical texturizing services?

Hair density measures the number of hairs per square inch and indicates how thick or thin the hair is. Think of walking through a forest that has many trees—the forest is very “dense” and it is difficult to walk through. On the other hand, think about walking across a field with very few trees—the field is not very dense. Hair density is important to consider when performing chemical texture services because it helps determine the number of blockings or subsections that will be best for the service that is performed. For example, in permanent waving, smaller blocks (subsections) and larger rods may be required for thick density hair. If the hair density is thin, however, smaller-diameter rods are required to form a good wave pattern close to the head. Avoid large blockings on thin hair growth in permanent waves as the strain may cause breakage. In chemical relaxing, subsections may need to be very small according to the density for proper saturation. Hair density can also indicate the amount of product that will be needed. If a client has a lot of hair, more product may be needed.

HAIR LENGTH

How does hair length affect chemical texturizing services?

In permanent waving and curl reformation processes, hair length may determine the wrapping technique to use. For example, while hair of average length may be wrapped normally, hair that is 6 inches or longer may not sit close enough to the scalp to develop a good, strong wave pattern in that area. In addition, the weight of long hair may relax or stretch the curl to the extent that the majority of the wave pattern remains mid-shaft and on the hair ends with very little lift in the scalp area. Like hair density, the length of the hair will also help determine the amount of product that will be needed for the chemical texturizing process. For example, clients with long hair may require additional waving solution, relaxer, or neutralizer to complete the process.

HAIR GROWTH PATTERN

How does hair growth pattern affect chemical texturizing services?

The direction of the natural hair growth creates hair streams, whorls, and cowlicks that can influence the finished style and reappear as new

growth occurs. The direction of hair growth must be considered when selecting the wrapping pattern and rod placement for permanent waves or curl reformations and for the direction of combing and smoothing when using chemical relaxers. When wrapping a permanent wave or curl reformation, wrapping in the opposite direction of a strong growth pattern may cause breakage.

Understand the Chemistry of Chemical Texture Services

After reading this section, you will be able to:

- LO 4** Describe how the ingredients in permanent waves, relaxers, and curl reformation services are chemically similar and chemically different from each other.
- LO 5** Explain the physical and chemical actions of permanent waving, chemical relaxing, and curl reformation processes.

Chemical texture services create permanent changes in the structure and appearance of the hair. As you learned earlier, hair is composed of three layers: cuticle, cortex, and medulla.

- *Medulla*: Because the medulla is considered to be empty space and may be present only in medium to coarse hair types, the cuticle and cortex are the two layers most affected by chemical texture services.
- *Cuticle*: The cuticle is the tough, outermost layer that protects the cortex from damage. The degree to which hair is resistant to chemical changes depends on the strength of the cuticle. The alkaline solutions and substances used in chemical texture services soften and swell the cuticle, allowing for penetration into the cortex (Figure 17-4).
- *Cortex*: The cortex gives the hair its strength, flexibility, elasticity, and shape. These characteristics are derived from the millions of polypeptide chains found in the keratin that makes up the cortex of the hair.

Restructuring keratin and polypeptide chains in the cortex: As you learned earlier, amino acids form proteins, and chemical reactions among proteins produce peptide linkages.

Hair develops and maintains its natural form by means of the physical and chemical cross-bonds in the cortex layer. The physical bonds, hydrogen and salt, are weaker bonds and are easily broken by shampooing and rinsing. Chemical bonds (disulfide) are broken or rearranged through the chemicals used in permanent waving, curl reformations, and relaxers.

figure 17-4
The cuticle layer.



Alkaline substances used in chemical texture products break the chemical bonds and allow for the softening and expansion of the hair. During this process, cystine (formed by disulfide or sulfur bonds) is altered slightly to become cysteine. Cysteine is an amino acid obtained by the reduction of cystine. This chemical action is important because it changes the chemical rearrangement of the inner structure of the hair as it assumes a new shape and form. For example, in permanent waving and curl reformation, the hair will take the shape and curl of the perm rod. In relaxing, the naturally curly hair is “relaxed” into a straighter, less curly form as it is manually “smoothed.”

After the hair has assumed the desired shape (curly on a perm rod or smoothed into a straighter form), it must be neutralized so that the hydrogen and sulfur cross-bonds in the cortical layer are permanently reformed. Cysteine is changed back to the cystine state during the process of oxidation and neutralization, which hardens the S bonds of the hair into the newly constructed shape. Both physical and chemical actions cause the bonds within the cortex to be rearranged and restructured during chemical texturizing services.

DEFINE THE PRINCIPLE ACTIONS OF CHEMICAL TEXTURIZING SERVICES

Chemical texture services involve two principal actions on the hair: physical and chemical (Figures 17-5 to 17-8). These actions are compared in Table 17-1.

figure 17-5
Polypeptide chains are formed when amino acids link together.



figure 17-6
Keratin proteins are long, coiled peptide chains.

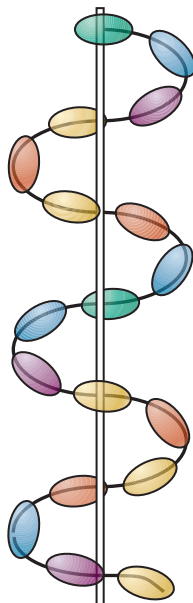


figure 17-7
Side bonds cross-link polypeptide chains together.

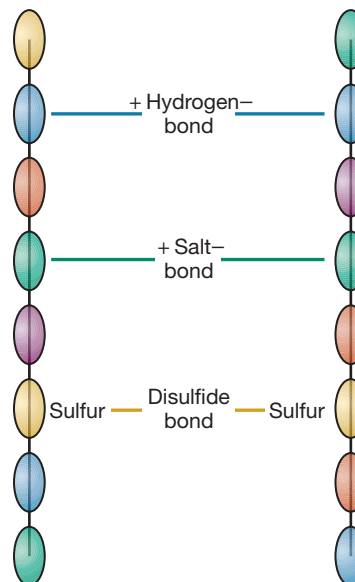


figure 17-8
A correct permanent wave service only alters the side bonds.



table 17-1

PHYSICAL AND CHEMICAL ACTIONS OF CHEMICAL TEXTURE SERVICES

Actions	Permanent Waving	Chemical Relaxing	Curl Reformation
Physical	- shampooing - rinsing - wrapping the hair around perm rods	- combing or smoothing relaxer through the hair - shampooing - rinsing - conditioning	- combing or smoothing rearranger through the hair for partial relaxation of the natural curl - rinsing - wrapping the hair on rods
Chemical	- Waving lotion/solution: - first, this softens and swells the cuticle layer - second, it breaks the disulfide bonds, called <i>reduction</i> - Neutralizer: - rehardens the newly arranged disulfide bonds - neutralizes any remaining waving lotion in the hair through oxidation	Relaxer is either thio based or sodium hydroxide based: <i>Thio relaxers:</i> - Thio relaxing products also require the use of a neutralizer to chemically oxidize the hair. - The reducing agent has a higher strength than that used in permanent waving. Most thio relaxers (THY-oh re-LAX-uhz) have a pH above 10 and are manufactured in cream form for better adhesion and control. - Relaxer softens and swells the hair and breaks apart the disulfide bonds. This action permits the removal of curl from the hair as the bonds are rearranged into a straighter position through the physical actions of combing and smoothing the hair. - The hair is then rinsed and neutralized with an oxidizing agent, such as hydrogen peroxide, that rebuilds the disulfide bonds broken by the relaxer. <i>Hydroxide relaxers:</i> - These are strong alkalis that can swell the hair up to twice its normal diameter. Hydroxide relaxers (hy-DRAHKS-yd re-LAX-uhz) can have a pH as high as 13.5. At these high concentrations, the hydroxide product permanently breaks the disulfide bonds to the point where they can never be reformed. This process is known as lanthionization (lan-THÉE-ohn-iz-ay-shun) and occurs as the disulfide bonds are converted to lanthionine bonds when the relaxer is rinsed and the hair is still at a high pH level. - Hydroxide relaxers are <i>not</i> compatible with thio relaxers because they use a different chemistry. - Hydroxide relaxing products are neutralized through the physical actions of the shampooing and rinsing process because the disulfide bonds that have been broken by this type of relaxer cannot be reformed through oxidation.	- Rearranger: - relaxes the natural curl by softening and swelling the cuticle, allowing for penetration into the cortex - ammonium thioglycolate in cream form (thicker consistency) - combed through the hair - Waving lotion (booster): - ammonium thioglycolate in lotion form (thinner consistency) - applied as hair is wrapped on perm rods - Neutralizer: - rehardens newly arranged disulfide bonds - neutralizes any remaining waving lotion in the hair through oxidation

Know about Permanent Waves

After reading this section, you will be able to:

- LO 6 Identify types of perm rods and end wrapping techniques.
- LO 7 Define *on-base*, *half off-base*, and *off-base* rod placement.

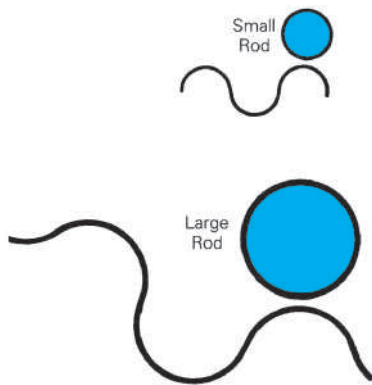
Permanent waving is a process that involves two principal actions on the hair:

1. *physical* action of wrapping the hair on perm rods
2. *chemical* changes caused by the waving solution and neutralizer

Permanent waves are performed on hair that has been freshly shampooed and is uniformly damp. A spray water bottle must be used to maintain even wetness throughout the hair during the wrapping procedure.

figure 17-9

The size of the rod determines the size of the curl.



UNDERSTAND THE CHEMICAL PERM WRAP

Perm Rods

In permanent waving, the size, shape, and type of curl are determined by the size, shape, and type of perm rod and the method used to wrap the hair. The size and shape of the rod determines the size of the curl (Figure 17-9).

The proper selection of perm rods is essential for successful permanent waving. The most commonly used rods are made of plastic.

Perm Rod Diameter

- The diameter of the rod controls the size of the curl.
- These diameters usually vary from 1/8 inch to 1 1/2 inches.
- The hair used in the subsection must be wrapped at least 1 1/2 times around the rod to create a curl formation.

Perm Rod Lengths

- Rods are available in lengths that typically measure from 1 3/4 inches to 3 1/2 inches.
- Perm rod lengths are known as:
 - short
 - medium
 - long

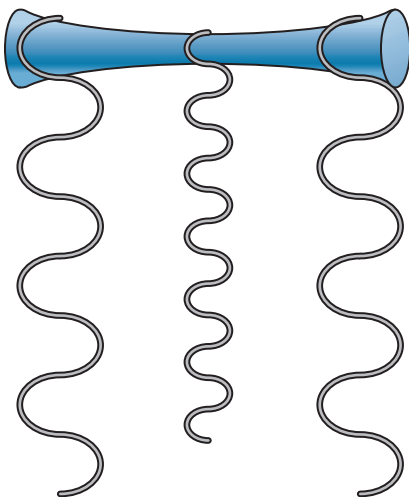
Perm Rod Shapes

Concave rods

- Concave rods are the most commonly used perm rod.
- Concave rods have a smaller diameter in the center, which gradually increases to a larger diameter at both ends.
- This shape produces a tighter curl in the center and a larger curl on the sides of the hair parting resulting in a more natural wave (Figure 17-10).
- Concave rods are used when a definite wave pattern, close to the head, is desired.

figure 17-10

Concave rod and resulting curl.



Straight rods

- Straight rods have a uniform circumference and diameter along the rod's length.
- This type of rod creates a consistently sized wave from one side of the hair parting to the other (Figure 17-11).
- Large, straight rods are usually used for a body wave that serves as a foundation for further styling.

Bender rods

- Bender rods are made of stiff wires covered by soft foam that permits bending into a variety of shapes.
- These rods measure about 12 inches in length and have a uniform diameter (Figure 17-12).

Circle or loop rods

- The circle tool or loop rod is a plastic-coated tool that also measures about 12 inches with a uniform diameter along the length of the rod.
- These rods are secured by fastening the ends together to form a circle (Figure 17-13).

End Papers or End Wraps

- **End wraps** are absorbent papers used to control the ends of the hair when wrapping and winding the hair on perm rods.
- End papers should extend beyond the ends of the hair to keep them smooth and straight, and to prevent “fishhooks.” A fishhook is a flaw in the wrapping that results in the tip of the hair bending in the opposite direction to that of the rest of the curl.
- End papers are especially effective in helping to smooth out the wrapping of uneven hair lengths. For very uneven lengths, you may use multiple end wraps up the hair strand to control the hair.

End Paper Wrapping Techniques

The most common end paper-wrapping techniques are the bookend wrap, the single flat wrap, and the double flat wrap.

- The **bookend wrap** uses one paper folded in half over the ends of the hair (Figure 17-14). It can be used with short rods or short lengths of hair. Be careful to distribute the hair evenly over the entire length of the rod and avoid bunching the ends together toward the center of the rod.
- The **single flat** or **single end wrap** uses one paper placed over the top of the hair parting being wrapped (Figure 17-15). Because there is paper on only side of the hair, it may be difficult to control the hair using this method.
- The **double flat** or **double-end wrap** uses two end papers, one placed under and one over the parting of hair being wrapped. Both papers should extend beyond the hair ends (Figure 17-16).

figure 17-11

Straight rod and resulting curl.

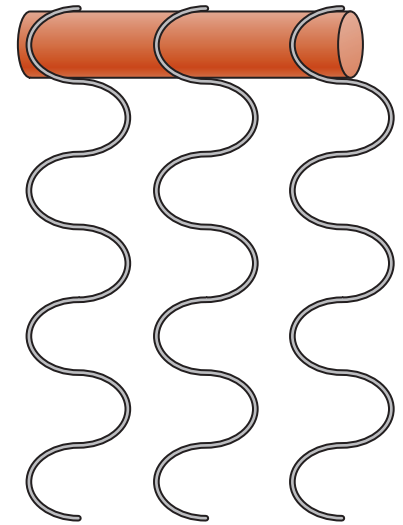


figure 17-12

Spiral wrapping with soft bender rods.



figure 17-13

Circle tools.



figure 17-14
Bookend wrap rodding a reformation curl.



figure 17-15
Single flat wrap rodding a reformation curl.



figure 17-16
Double-end wrap rodding a reformation curl.



Bookend Wrapping Mini-Procedure

1. Part and comb subsection up and out until all the hair is evenly directed and distributed (see **Figure 17-17**).
2. Hold the subsection between the index and middle fingers. Fold and place the end paper over the subsection, forming an envelope (see **Figure 17-18**).
3. Hold the subsection smoothly and evenly. Slide the paper envelope a small fraction beyond the hair ends (see **Figure 17-19**).
4. With your right hand, pick up the rod (see **Figure 17-20**).

figure 17-17
Bookend wrap step 1.

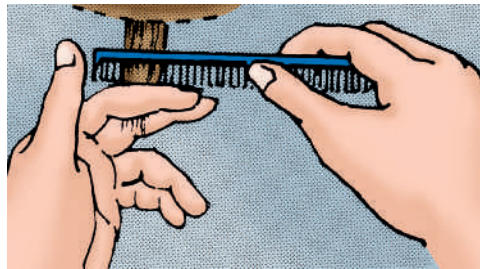


figure 17-18
Bookend wrap step 2.

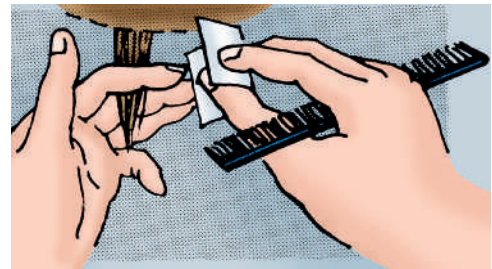


figure 17-19
Bookend wrap step 3.

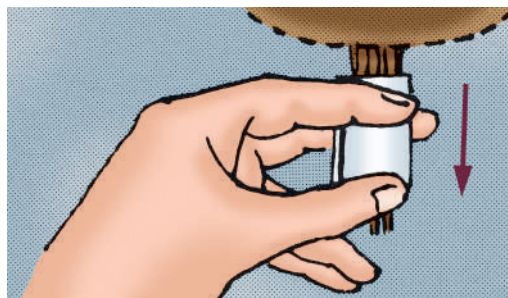


figure 17-20
Bookend wrap step 4.

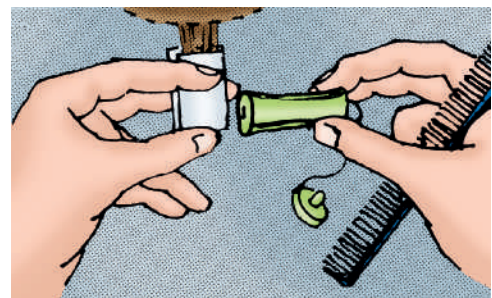


figure 17-21
Bookend wrap step 5.

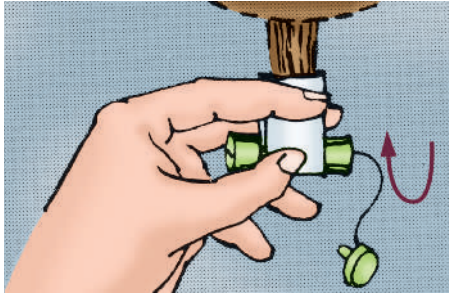


figure 17-22
Bookend wrap step 6.

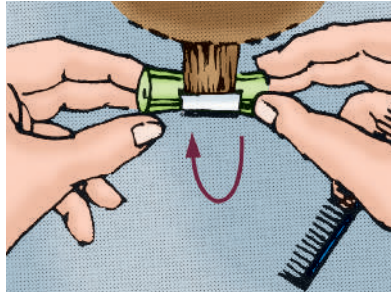
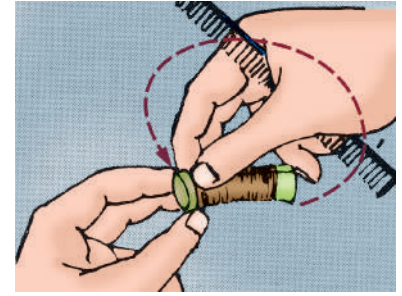


figure 17-23
Bookend wrap step 7.



5. Place the rod under the folded end paper, parallel to the parting. Draw the end paper and rod toward the hair ends until they are visible above the rod, and start winding the end paper and hair under toward the scalp (see **Figure 17-21**).
6. Wind the hair smoothly, without tension, on the rod (see **Figure 17-22**).
7. Fasten the band evenly across the wound hair at the top of the rod (see **Figure 17-23**).

Single Flat Wrap Mini-Procedure

1. Place the end paper on top of the subsection and hold it flat to prevent bunching (see **Figure 17-24**).
2. Place the rod under the subsection, holding it parallel with the parting. Then, draw the end paper and rod downward until the hair ends are covered (see **Figure 17-25**).
3. Roll the end paper and subsection under, using the thumb of each hand to keep the strands smooth. Wind the subsection on the rod to the scalp without tension. Fasten the band across the top of the rod in the same manner as the bookend paper wrap (see **Figure 17-26**).

figure 17-24
Single flat wrap step 1.

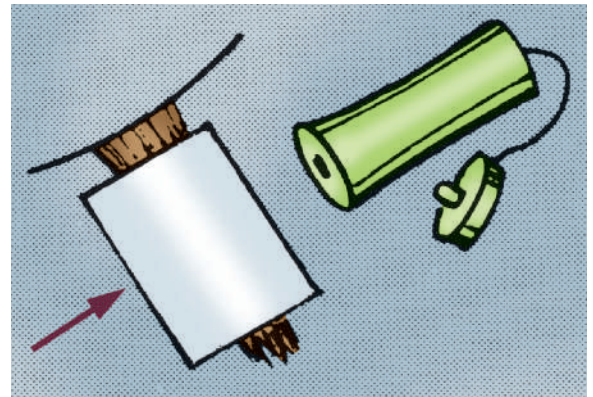


figure 17-25
Single flat wrap step 2.

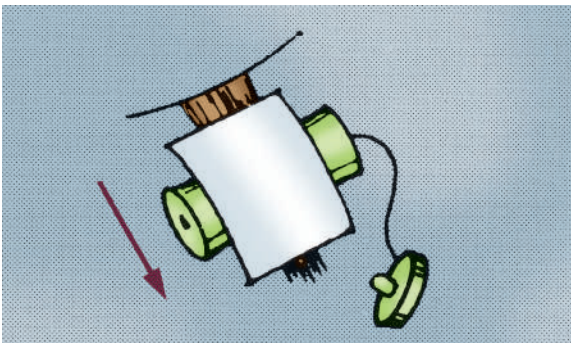
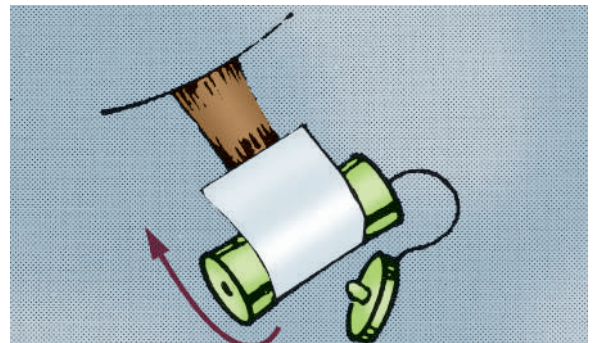


figure 17-26
Single flat wrap step 3.



Double Flat Wrap Mini-Procedure

figure 17-27

Double flat wrap step 1.

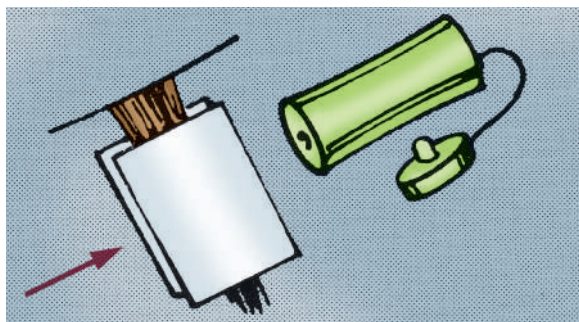
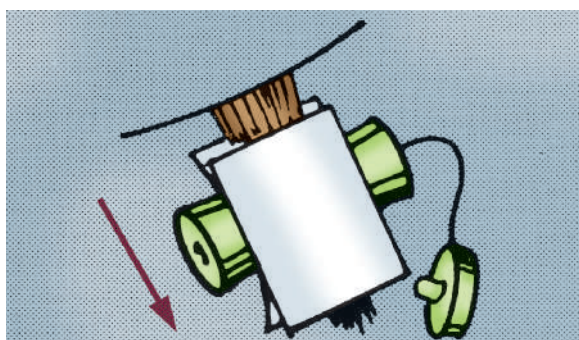


figure 17-28

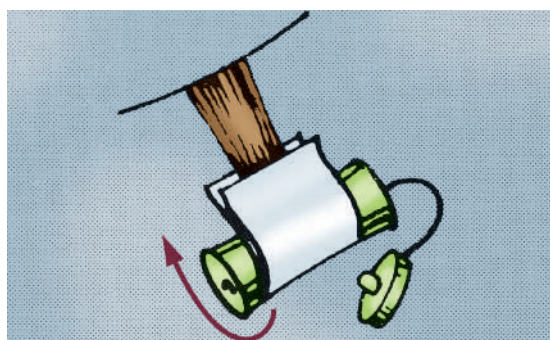
Double flat wrap step 2.



1. Place one end paper beneath the hair subsection and the other on top (see Figure 17-27).
2. Place the rod under double-end papers, parallel with the parting. Draw both papers toward the hair ends (see Figure 17-28).
3. Wind the subsection smoothly on the rod to the scalp without tension. Then fasten the band across the top of the rod in the same manner as the bookend wrap (see Figure 17-29).

figure 17-29

Double flat wrap step 3.



The *elastic band* secures the hair and the rod into the desired position to prevent the curl from unwinding during the procedure.

Sectioning

Perm wraps begin with sectioning the hair into panels. The size, shape, and direction of these panels will vary depending on the wrapping pattern and the type of tool being used. Each panel is further divided into subsections called **base sections**. The base sections are actually partings taken from the subsection that measure *almost* the same length and width as the rod or perming tool.

The number of base sections and the rod sizes will vary with each client depending on the sectioning pattern used and the texture, density, and elasticity of the hair (Table 17-2). The average base section, or parting, should be slightly shorter than the length of the rod. If the rod is shorter than the length of the blocking, the hair will slip off the ends of the rod during winding and result in an uneven curl formation.

The texture, elasticity, porosity, and condition of the client's hair all help determine how the hair should be sectioned and subsectioned, which rods to use, and where the application of waving solution should begin. Be guided by your instructor.

figure 17-30
On-base placement.



figure 17-31
Half off-base placement.



figure 17-32
Off-base placement.



table 17-2
SUGGESTED BASE SECTIONS AND ROD SIZES

Texture	Density	Elasticity	Base Section	Diameter of Rod
Coarse	Thick	Good	Narrow	Large
Medium	Average	Normal	Smaller	Medium
Fine	Thin	Poor	Smaller	Small
Damaged	Average	Very Poor	Smaller	Large

Base Control

Base control refers to the position of the perm rod or tool in relation to its base section and is determined by the angle at which the hair is wrapped. Rods can be wound *on-base*, *half off-base* (*half-base*), or *off-base*:

- In *on-base* placement, the hair is projected about 45 degrees beyond perpendicular (90 degrees) to its base section and the rod is placed *on* the base section (Figure 17-30). On-base placement results in greater volume at the scalp area and maximum movement.
- *Half off-base* placement results from wrapping the hair at an angle of 90 degrees to its base section (Figure 17-31). At this elevation of the hair, the rod is positioned half off its base section. Half off-base placement results in medium volume and movement.
- *Off-base* placement is achieved by wrapping the hair at a 45-degree angle below perpendicular to its base section (Figure 17-32). The rod is positioned completely off its base section and creates the least amount of volume and movement in the style.

Base Direction

Base direction refers to the directional pattern in which the hair is wrapped. Although wrapping with the natural direction of hair growth causes the least

figure 17-33
Vertical base direction.

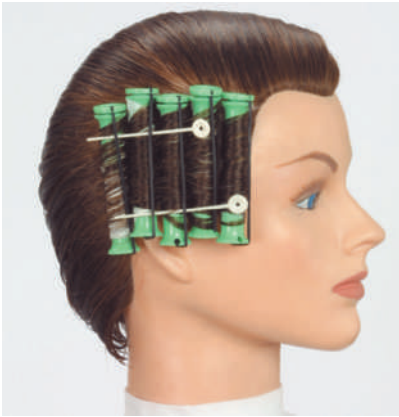


figure 17-34
Horizontal base direction.



figure 17-35
Croquignole perm wrap.

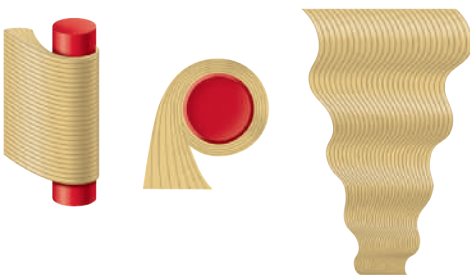
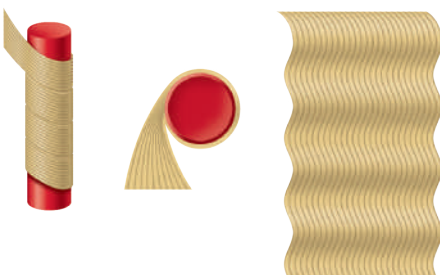


figure 17-36
Spiral perm wrap.



amount of stress on the hair, wrapping forward, backward, or to the side can create special effects in the final hairstyle. Be careful not to wrap the hair opposite a strong growth pattern as it may result in breakage.

Base direction also refers to the horizontal, vertical, or diagonal partings and positioning of the rod on the head, as pictured in **Figures 17-33** and **17-34**.

Wrapping/Rodding Techniques

The two basic methods used to wrap the hair around the perm rod are:

Croquignole

When using the **croquignole perm wrap** (KROH-ken-yohl) technique, the hair is wound from the hair ends toward the scalp. This overlapping of the hair layers with each revolution of the rod increases the size of the curl as it nears the scalp area, with a tighter curl at the ends (**Figure 17-35**).

Spiral

- A spiral wrap can be accomplished in two ways:
 - from the hair ends toward the scalp, or
 - from the scalp toward the hair ends.

In both cases, the rod or perm rod is positioned vertically as the hair parting is “rocked” or spiraled along the length of the rod. Straight rods are usually used for a spiral wrap and tend to produce a more uniform curl when the hair is spiraled properly along the rod (**Figure 17-36**). A spiral wrap is performed at an angle that positions the hair in a spiral pattern along the length of the rod. This technique produces a uniform curl formation from the scalp to the ends and is especially appropriate for long hair designs.

To form a uniform wave, the hair must be wrapped smoothly and neatly on each rod without stretching. The hair should not be stretched or wrapped too tightly because that could interfere with the penetration and expansion process of the waving solution and the contraction process of the neutralizer. Tight wrapping can also result in hair breakage due to the amount of tension put on the hair.

Water versus Lotion Wrapping

Water Wrap

A **water wrap** is the action of wrapping the hair around rods when it is moist with water. It is important to keep all of the hair moist throughout the wrapping process. If some of the hair has dried, remoisten it using a spray bottle. Although most perms are wrapped in this manner, always follow the manufacturer’s directions.

Lotion Wrap

A **lotion wrap** is the application of permanent wave solution to one working panel or section of hair just before wrapping the hair on rods

to pre-soften resistant hair. This technique is usually used with alkaline (cold) waves to make it easier to wrap resistant, straight hair types—it “jump starts” the processing action. Once the hair is wrapped, the remaining solution is applied to the entire head. Always follow the manufacturer’s instructions.

Permanent Waving Wrapping Patterns

Once the client consultation has been completed, the barber should have a definite plan of action designed to achieve the desired results. This plan should include the selection of the permanent waving product, rod size(s), and wrapping pattern that will be used for the service. The parting technique most often used is a straight part. However, zigzag partings, known as the *weave technique*, can be used for an entire perm or to create smooth transitions from wrapped areas to non-wrapped areas within a partial perm.

There are four common wrapping patterns used in permanent waving:

- *Basic wrap*

In this wrapping pattern, all the rods within a panel are positioned in the same direction on equal-size bases (Figure 17-37). All base sections are horizontal and slightly smaller than the length of the rod and even to the width of the rod. The base control is usually half off-base, although other bases may be used.

- *Curvature perm wrap*

This wrapping pattern is one of the best to use for men’s styles as it produces a more natural-looking wave pattern. The movement curves within sectioned-out panels with partings and bases following the natural curvature of the head and distribution of the hair (Figure 17-38).

- *Bricklay perm wrap*

The base sections are offset from each other row by row to prevent noticeable splits in the hair (Figure 17-39). The bricklay pattern will blend hair from one area to another and may be preferred for men’s styling over the basic perm wrap pattern.

- *Piggyback perm wrap*

The piggyback perm wrap uses two perm rods for each parting of hair for even curl formation on long and/or dense hair. The first rod is placed about midway between the scalp and hair ends. This section is wrapped toward the scalp and leaves a tail of hair that extends from the mid-shaft point to the hair ends. The “tail” is then wrapped from the ends to the midway point and secured and positioned across the first rod (Figure 17-40). In this way, the second rod is “piggybacking” the first rod.

figure 17-37

Basic perm wrapping pattern.



figure 17-38

Curvature perm wrapping pattern.



figure 17-39

Bricklay perm wrapping pattern.

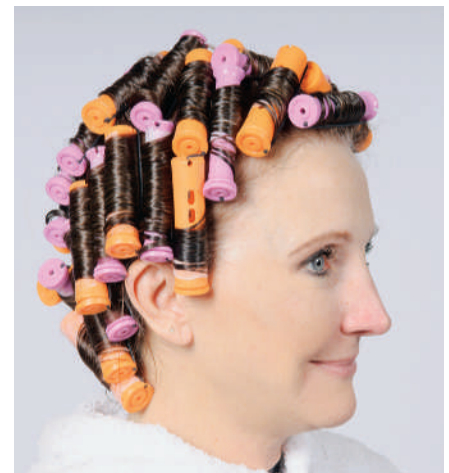


figure 17-40
Piggyback wrap.



TYPES OF PERMANENT WAVES

Permanent wave chemistry is constantly being refined and improved. Different formulas designed for a wide variety of hair types are available from today's manufacturers. Waving lotions and neutralizers for both acid-balanced and alkaline perms are available with new conditioners, proteins, and natural ingredients that help protect and condition the hair during and after perming.

Stop-action processing is incorporated into many waving lotions to ensure optimum curl development. The curling takes place within a fixed time, without the risk of overprocessing or damaging the hair. Special pre-wrapping lotions have also been developed to compensate for hair with varying degrees of porosity.

Alkaline Perms (Cold Waves)

The first **alkaline or cold waves** (AL-kuh-line WAYVZ) were developed in 1941. The main active ingredient or reducing agent in alkaline perms is **ammonium thioglycolate (ATG)** (uh-MOH-nee-um THY-oh-GLY-kuh-layt). This chemical compound is made up of ammonia and thioglycolic acid. The pH of alkaline waving lotions generally falls within the range of 9.0 to 9.6, depending on the amount of ammonia. These high pH levels cause the hair shaft to swell and the cuticle layer to lift, allowing penetration into the cortex. This permanent wave may require a plastic cap for processing. Always read the manufacturer's directions carefully before proceeding with the service.

The benefits associated with alkaline perms are strong curl patterns, fast processing time (5 to 20 minutes), and room temperature processing. Alkaline perms are generally used when perming resistant hair types, when a strong or tight curl is desired, or when the client has a history of early curl relaxation.

REMINDER

Repeated exposure to GMTG has been known to cause allergic sensitivity in practitioners and clients. Watch clients closely for signs of allergy or irritation. If problems develop, suggest they see a dermatologist.

True Acid Waves

True acid waves (TROO AS-ud WAYVZ) were first introduced in the early 1970s. These perms have a pH range between 4.5 and 7.0 and use **glyceryl monothioglycolate (GMTG)** (GLIS-ur-il mon-oh-THY-oh-GLY-kuh-layt) as the primary reducing agent. This lower pH is gentler on the hair and typically produces a softer curl and less damage than alkaline cold waves. Most true acid waves require the addition of heat from a hair dryer to accelerate the processing and are considered to be *endothermic* perm products. **Endothermic waves** (en-duh-THUR-mik WAYVZ) require the use of an outside heat source to activate chemical reactions and processing. It should be noted that although a pH of 7.0 is neutral on the pH scale, a pH of 5.0 is neutral for hair. Since every step in the pH scale represents a 10-fold change in pH, a pH of 7.0 is 100 times more alkaline than the pH of hair. Even pure water will cause the hair to swell and expand.

- The benefits of true acid waves are:
 - softer curl patterns
 - slower, but more controllable processing time (usually 15 to 25 minutes)
 - gentler treatment for delicate hair types
- Generally, true acid perms should be used when:
 - “perming” is sufficient delicate, fragile, or color-treated hair
 - a soft, natural curl or wave pattern is desired
 - style support, rather than strong curl, is desired

Acid-Balanced Waves

Most of today’s acid waves have a pH between 7.8 and 8.2. Modern acid waves are **acid-balanced** waves that have a 7.0 or neutral pH. Acid-balanced waves process at room temperature and do not require the heat of a hair dryer for processing. Although GMTG is the primary reducing agent in acid-balanced waves, these waving products usually contain some ATG.

- All acid-balanced waves have three chemicals:
 - permanent waving solution
 - activator
 - neutralizer
- The permanent waving solution usually contains ATG and the activator GMTG. These products must be added together just prior to use.
- Acid-balanced waves process more quickly and produce firmer curls than true acid waves.



CAUTION

Accidentally mixing the contents of the activator tube with the neutralizer instead of the permanent waving solution will cause a violent chemical reaction that can cause injury, especially to the eyes.

Exothermic Waves

Like acid-balanced perms, all **exothermic waves** (ek-soh-THUR-mik WAYVZ) have three components:

- permanent waving solution
- activator
- neutralizer

The activator in an exothermic wave, however, contains an oxidizing agent (usually hydrogen peroxide) that causes a rapid release of heat when mixed with the waving solution. The chemical reaction that releases heat and causes the waving solution to become warm can usually be felt when holding the bottle of solution. This rise in temperature increases the rate of the chemical reaction, which in turn shortens the processing time.

Ammonia-Free Waves

Ammonia-free waves use *alkanolamines* to replace ammonia and are gaining in popularity because of their low odor. Ammonia-free does not mean free of damage, so although these new products may not smell as strong as ammonia, they can still have a high alkaline pH. Processing takes place at a lower pH. Alkanolamines evaporate slowly, which is why they do not give

off odors, and maintain the same pH level throughout the entire processing time. This constant pH level can ultimately be more damaging to the hair than are waves containing ammonia.

Thio-Free Waves

Thio-free waves use *cysteamine* or *mercaptamine* as the primary reducing agent. When used at high concentrations, these thio compounds with a pH range of 7.0 to 9.6 can be just as damaging as regular thio formulations.

Low-pH Waves

Sulfates, sulfites, and bisulfites offer an alternative to ATG but are weak and do not produce a firm curl. Sulfite perms are usually marketed as *body waves* and have a pH range of 6.5 to 7.0.

Strengths of Waving Solutions

Although the strength of a waving solution can be adjusted by increasing the amount of the active ingredient, today's wide range of waving products virtually eliminates the need to do so. Be cautioned that if the strength of a solution is adjusted through the addition of ammonia, the pH level should not exceed 9.6.

Most manufacturers market three or more strengths of permanent waving products:

- *Mild*: for damaged, porous, or tinted hair
- *Normal*: for normal hair with good porosity
- *Resistant*: for resistant hair with less porosity

Pre-wraps

Some permanent waving product packages contain a **pre-wrap solution** that is applied to the hair before wrapping on rods. In most cases, the pre-wrap is simply a leave-in conditioner that helps equalize the porosity of the hair to ensure even penetration of the waving lotion and to protect the hair from unnecessary damage. Always follow the manufacturer's directions.

Perm Selection

For a successful permanent waving service, it is essential to select the proper waving product for the client's hair type and desired result. After a thorough client consultation, the barber should be able to determine which type of permanent waving product is best suited to the client's hair condition. Most resistant hair types require an alkaline wave; alkaline or acid-balanced perms can be used on most normal hair types; and acid-balanced or true acid perm formulas are the best choice for most tinted, highlighted, or delicate hair types. Use **Table 17-3** as a general guide to the selection of the most common types of permanent waves.

table 17-3

PERMANENT WAVE SELECTION

Perm Type	Active Ingredient	Wrapping Method	Process	Recommended Hair Type	Results	Advantages	Disadvantages
Alkaline/cold wave pH: 9.0 to 9.6	Ammonium thioglycolate (ATG)	Lotion wrap or water wrap	Room temperature	Coarse, thick, or resistant	Firm, strong curls	Processes quickly at room temperature	Unpleasant ammonia odor; may damage delicate hair
Exothermic wave pH: 9.0 to 9.6	Ammonium thioglycolate (ATG)	Water wrap	Exothermic	Coarse, thick, or resistant	Firm, strong curls	Faster processing time	Unpleasant ammonia odor; may damage delicate hair
True acid wave pH: 4.5 to 7.0	Glyceryl monothioglycolate (GMTG)	Water wrap	Endothermic	Extremely porous or very damaged	Soft, weak curls	Low pH produces minimal swelling	Requires heat from hair dryer; will not produce firm, strong curls
Acid-balanced wave pH: 7.8 to 8.2	Glyceryl monothioglycolate (GMTG)	Water wrap	Room temperature	Porous or damaged	Soft curls	Minimal swelling; processes at room temperature	Repeated exposure may cause allergic sensitivity in clients and stylists
Ammonia-free wave pH: 7.0 to 9.6	Monoethanolamine (MEA) or aminomethylpropanol (AMP)	Water wrap	Room temperature	Porous to normal	Medium to fine curls	No unpleasant ammonia odor	Overall strength varies with different manufacturers
Thio-free wave pH: 7.0 to 9.6	Mercaptamine or cysteamine	Water wrap	Room temperature	Porous to normal	Medium to fine curls	May be gentler, depending on formula	Overall strength varies with different manufacturers
Low-pH wave pH: 6.5 to 7.0	Ammonium sulfite or ammonium bisulfite	Water wrap	Endothermic	Normal, fine, or damaged	Weak curl or body wave	Minimal swelling	Requires heat from hair dryer; produces weak curls

figure 17-41
S pattern.



figure 17-42
Underprocessed hair.



figure 17-43
Overprocessed hair.



PERMANENT WAVE PROCESSING AND WAVE FORMATION

The strength of any permanent wave is based on the concentration of its reducing agent. In turn, the amount of processing time is determined by the strength of the permanent waving solution and the porosity level of the hair. Most of the processing takes place within the first 5 to 10 minutes, so the amount of processing should be determined by the strength of the solution and not by how long the perm processes. If the hair is not sufficiently processed after 10 minutes, it may require reapplication of the solution. The next time the client receives a perm service, a stronger solution may be used. Conversely, if the client's hair has been overprocessed, a weaker solution should have been used.

As the hair is processing, the waves form a deep-ridged pattern. The wave has reached its peak when it forms a firm letter S shape (Figure 17-41). The S pattern reaches a desirable peak only once during the perm process. Shortly after the S is well formed, the hair may become frizzy unless processing is stopped. Frizziness indicates that the processing time has reached its absolute maximum. Beyond this point, the hair becomes overprocessed and damaged.

Different conditions and hair textures will cause the quality of wave patterns to vary. Hair of good texture will show a firm, strong pattern, whereas hair that is weak or fine will not produce a firm pattern.

Underprocessing

- This is when hair has not been sufficiently softened to permit the breaking and rearrangement of the disulfide bonds.
- This results in a limp or weak wave formation with undefined ridges within the S pattern.
- The hair retains little or no wave formation and is unable to hold the desired curl (Figure 17-42).
- Reapplication of the permanent waving solution is necessary to complete the process.

Overprocessing

- The hair can also appear to be too weak to hold a curl (Figure 17-43).
- Any solution that can process the hair properly can also overprocess it.
- Solution left on the hair too long results in overprocessing and if too many disulfide bonds are broken, the hair may not have enough strength to hold the desired curl.
- Signs that the hair has been overprocessed include:
 - a weak curl formation
 - a very curly appearance when wet but completely frizzy when dry
 - an inability to be combed into a suitable wave pattern
- In hair that has been overprocessed, the elasticity of the hair has been damaged to the point where it is unable to contract into the wave formation.
- Overprocessed hair usually feels harsh after being dried and should be given reconditioning treatments.

Test Curls

Test curls help determine how the client's hair will react to the permanent waving process. A test curl provides information about how to obtain the best possible results out of the perm service. Test curls are performed by wrapping one parting of hair in three different areas of the head: the top, side, and nape (Figure 17-44).

Test curls enable the barber to observe the following aspects of the hair:

- speed of wave formation
- degree of wave formation
- exact time when peak of wave formation is reached
- identification of resistant areas
- appropriateness of product selection

figure 17-44

Applying wave lotion to test curls.



P 17-1 Preliminary Test Curl for a Permanent Wave pages 612-613

NEUTRALIZATION

Permanent waving neutralizers are actually oxidizers that stop the action of permanent wave solutions and rebuild the disulfide bonds broken during processing. This process is known as **neutralization** (noo-truh-ly-ZAY-shun). The most common neutralizer is hydrogen peroxide with a concentration range between 5 volume (1.5 percent) and 10 volume (3 percent). Other types of neutralizers are sodium bromate and sodium perborate.

Neutralization has two important functions:

1. Rebuild the disulfide bonds that were broken and rearranged by the waving solution.
2. Deactivate any waving solution that remains in the hair after rinsing.

If the hair is not properly neutralized, the curl will relax or straighten within one to two shampoos. As with waving solutions, there may be slightly different procedures recommended for individual products. To achieve the best possible results, always read the directions carefully.

Proper rinsing is key to a perm. The rinsing process must last for at least 5 minutes, adding an additional minute for every inch the hair is longer than 5 inches. After thoroughly rinsing the permanent wave solution from the hair, it should be blotted until no excess water is absorbed into the towel. This is an important step in the neutralization process because excess water left in the hair prevents even saturation of the neutralizer and dilutes its properties.

Neutralizer Application

- The neutralizer should be applied to the top and bottom of each rod to assure saturation and even distribution of the product.
- Always give clients a towel to protect their eyes from excess or dripping neutralizer. A fresh cotton coil should also be used for additional safety and client comfort.

Rinsing the Neutralizer

Depending on the manufacturer's directions, neutralizers are rinsed from the hair in one of two ways:

- *Rinsing with the rods in place*
 - After a thorough rinsing with tepid water, followed by blotting, the rods are carefully removed in preparation for styling.
 - Some directions may require a second rinse after the removal of the rods.
- *Rinsing after the rods have been removed*
 - An alternative method is to carefully remove the rods, without stretching the hair, after neutralization has taken place.
 - Apply the balance of the neutralizing solution and allow it to remain on the hair for an additional minute.
 - Then rinse with tepid water and proceed with setting and/or styling the hair.

Post-permanent Wave Care

Most manufacturers recommend a 24- to 48-hour waiting period before shampooing freshly permed hair. Always follow the manufacturer's directions and be sure to educate your clients about the most appropriate shampoos and conditioners to use after a perm service. Unless there are signs of scalp irritation, deposit-only haircolor can also be applied sooner than three days after a permanent wave.

Reconditioning treatments can be used in the aftercare and between permanent waves. Effective post-perm care helps keep the hair in the best possible condition until the next chemical service. Suggest the following guidelines to your clients:

- Shampoo the hair as needed with an acid-balanced shampoo.
- Use a moisturizing hair conditioner.
- Schedule regular shop visits for trims and/or conditioning treatments to maintain the hairstyle.

figure 17-45
Partial perm style.



PARTIAL PERMANENT WAVES

Partial perming means that only a section of the hair is curled. In men's permanent waving, this is usually hair on the top and crest area of the head (**Figure 17-45**). Partial perms can be used to create volume and lift in these areas or to re-curl previously curled sections that have been trimmed due to normal hair growth and maintenance haircuts.

The same wrapping patterns and techniques used for a full permanent wave can be used for a partial perm with the following considerations:

- To make a smooth transition from the wrapped section to the non-wrapped section, wrap as far as the client wants, then before neutralizing, remove the bottom row and soften the curl. This will transition from permed to straight hair.
- After wrapping, place a coil of cotton around the wrapped rods as well as around the entire hairline.

- Before applying the waving lotion, apply a heavy conditioner or barrier cream to the sections that will not be permed to protect this hair from the effects of the waving lotion.

Many clients need the added texture and fullness that only a permanent wave can give. A permanent wave can help temporarily overcome common hair problems by redirecting a cowlick, making limp or unmanageable hair easier to style, or making sparse hair look fuller. Although men and women's hairstyles may be different, the techniques for permanent waving are essentially the same.

SPECIAL PROBLEMS

Dry, Damaged Hair

Dry, brittle, damaged, or overporous hair should be given reconditioning treatments before a permanent wave service. Avoid any treatment requiring massage or heat as this could create a sensitive scalp and irritation when the waving solution or neutralizer is applied. Special fillers that contain protein are available for reconditioning the hair and helping equalize its porosity. Some fillers also contain lanolin and cholesterol, which may help protect the hair against the harshness of the permanent waving solution.

Tinted or Lightened Hair

Hair that has been tinted or lightened should be shampooed with an extra-mild shampoo before waving. It may be advisable to use a pre-wrap or other leave-in conditioning product. While pre-wraps and leave-in conditioners may be sufficient for stronger hair types that have been previously tinted or lightened, extremely porous hair will probably not benefit from these applications. Extremely porous hair may absorb too much conditioner, which in turn may interfere with waving solution penetration and curl formation. Always select permanent waving solutions that are formulated for tinted and lightened hair conditions and always take the time to perform test curls.

Hair Tinted with Metallic Dye

Some over-the-counter haircoloring products still use metallic dyes in the formulations. Hair tinted with a metallic dye must first be treated with a dye remover to avoid hair discoloration or breakage. Do not chemically wave the hair if the test curls break or discolor. This type of haircoloring product is difficult to remove so the best option is to cut the color-treated hair in a series of shop visits until all the color is removed. Then proceed with the perm service on virgin hair, followed by the application of professional haircoloring products.

Curl Reduction

Sometimes a client is not pleased after a permanent wave because the hair seems too curly. If the hair is fine, do not suggest curl reduction before shampooing two or three times. This type of hair relaxes to a greater extent than does normal or coarse hair. Usually, after the second shampoo, the hair relaxes to a satisfactory level. If the hair has a normal or coarse texture, curl reduction may be done immediately following neutralization or



CAUTION

Do not attempt curl reduction on hair that has been overprocessed, as such treatment will only damage the hair further.

after a few days. Permanent waving solution may be used to relax the curl where required. Carefully comb it through the hair to widen and loosen the wave. When sufficiently relaxed, the hair is rinsed, towel-blotted, and neutralized.

Air-Conditioning and Heating Units

Because of its cooling effects, air-conditioning will usually slow the action of permanent waving solutions and additional processing time may be required. Make sure that clients are seated in an area of the shop away from drafts, vents, and fans to avoid slowing the processing time. Conversely, sitting too close to a heating vent or hood hair dryer can speed up the processing time.

SAFETY PRECAUTIONS FOR PERMANENT WAVING

- Always protect the client's clothing with the proper waterproof cape.
- Use two towels, one under the cape and one over the cape.
- Always examine the client's scalp before a perm service. Do not proceed if abrasions are present.
- Do not proceed with the perm if the client has ever experienced an allergic reaction to the products.
- Do not perm excessively damaged hair or hair that has been treated with hydroxide relaxers.
- Always apply a protective cream barrier around the client's hairline before applying the waving solution.
- Immediately replace cotton coils or towels that have become saturated with solution.
- Always protect the client's eyes when applying waving and neutralizing solutions by providing the client with a clean towel to hold over the eyes during the application. In case of accidental exposure, rinse thoroughly with cool water.
- Always follow the manufacturer's directions.
- Do not dilute or add anything to waving or neutralizing solutions unless specified in the manufacturer's directions.
- Wear gloves when applying solutions.
- Do not save opened or unused products, as the strength and effectiveness will change if not used promptly.
- Unless otherwise specified in the product instructions, apply waving and neutralizing solutions liberally to the top and underside of each rod.
- Start at the crown and progress systematically down each section. (Some barbers prefer to start at the top of the head.) Be sure that the surface area of the wound hair is wet with lotion so saturation is even.
- Follow the same application pattern for the neutralizer as used with the waving solution to avoid missing any rods.

- Sometimes it is necessary to resaturate the rods during processing. This may be due to evaporation of the solution, dryness of the hair, hair that was poorly saturated the first time, improper selection of solution strength, or failure to follow the manufacturer's directions. Reapplying the solution will hasten processing, so watch the wave development closely as negligence may result in hair damage.
- Make sure the hair stays evenly wet while wrapping on rods.
- Have client remove jewelry to avoid discoloration or skin reactions.
- Frequently change towels as they become soiled or saturated.

Know about Chemical Hair Relaxers

After reading this section, you will be able to:

- LO8** Identify two types of chemical relaxers.
- LO9** Explain the difference between *base* and *no-base* relaxers.
- LO10** List three strand tests to be performed before a chemical relaxing process.

Chemical hair relaxing is the process of rearranging the basic structure of curly or wavy hair into a relaxed or straighter form. Like the permanent waving process, chemical relaxers change the shape of the hair by breaking and rearranging disulfide bonds in the hair. Relaxers can also soften coarse hair and make it more manageable.

The basic products used in the chemical relaxing process are:

- chemical relaxers (straighteners)
- neutralizers or neutralizing shampoos
- protective bases or creams
- conditioners

TYPES OF RELAXERS

The two most common types of relaxers are thio and hydroxide relaxers. Neither type of relaxer allows pre-shampooing unless there is an excessive buildup of dirt or styling products. In rare instances when the hair requires cleansing, the shampoo service should be performed gently to avoid scalp irritation and the hair should be thoroughly dried before the relaxer is applied.

Both types of relaxers are applied in the same manner, but thio relaxers require the application of a chemical neutralizing solution, whereas hydroxide relaxers are neutralized through the physical actions of rinsing and shampooing with a neutralizing shampoo product. Neutralizing shampoos



CAUTION

Applying a thio-based product on hair that has been previously treated with sodium hydroxide OR applying a hydroxide-based product on hair that has been previously treated with a thio-based product can cause extreme damage or hair breakage.

or normalizing lotions are basically acid-balanced shampoos that neutralize any remaining hydroxide in the hair and help lower the pH of the hair and scalp.

Thio Relaxers

Thio (ATG) is the same reducing agent used in permanent waving, but thio relaxers usually have a pH above 10.0 and a higher concentration of ATG than permanent wave products. Thio relaxers are also thicker in consistency than permanent waving lotions for better adhesion and product control.

Hydroxide Relaxers

Hydroxide relaxers are ionic compounds formed by a metal combined with oxygen and hydrogen. Metal hydroxide relaxers include sodium hydroxide, potassium hydroxide, lithium hydroxide, and guanidine hydroxide.

Sodium Hydroxide Relaxers

These are known as *lye* relaxers. Sodium hydroxide is the oldest and most commonly used chemical relaxer. Sodium hydroxide content varies from 5 percent to 10 percent, with a pH range of 12.5 to 13.5. Generally speaking, the higher the percentage of sodium hydroxide, the faster the chemical reaction on the hair. The higher the pH factor, the greater the danger of hair damage or breakage.

Potassium and Lithium Hydroxide Relaxers

These are often sold as *no-mix/no-lye* relaxers. Although these two relaxers are not technically lye products, their chemistry is identical and there is little difference in their performance.

Guanidine Hydroxide Relaxers

These are usually advertised as no-lye relaxers. These relaxers contain two products, a relaxer cream and an activator, that must be mixed just prior to use and are recommended for sensitive scalps.

Calcium Hydroxide Relaxers

Many *calcium hydroxide* relaxers are often mistakenly referred to as *no-lye* relaxers. Obviously, this type of marketing statement can be misleading, so care must be taken when choosing the relaxer product. Calcium hydroxide relaxers require the addition of an activator. The strength of the relaxer is determined by the amount of activator used in the mixture. Although these relaxers are considered to be mild and tend to work more slowly on the hair, there are many professionals who feel that calcium hydroxide relaxers are more damaging to the cuticle layer of the hair than other hydroxide relaxers.

“Base” versus “No-Base” Relaxers

Hydroxide relaxers are usually sold in *base* and *no-base* formulas.

- **Base relaxers** require the application of a base cream to the entire scalp prior to relaxer application.

- **No-base relaxers** contain a base cream that is designed to melt at body temperature and do not require the application of a separate protective base.

RELAXER STRENGTHS

Most chemical relaxers are available in a variety of different strengths: mild, regular, and super. The strength of the relaxer reflects the concentration of hydroxide in its formulation.

- Mild strengths are recommended for fine, color-treated, or damaged hair.
- Regular-strength relaxers are intended for normal hair textures.
- Super strengths should be used for maximum straightening on extremely curly, coarse hair.
- Use [Table 17-4](#) as a guide for selecting chemical relaxing products.

STRAND TESTS FOR CHEMICAL RELAXERS

Preliminary strand tests for chemical relaxer applications include:

- porosity test for determining the degree of porosity
- elasticity test for determining the degree of elasticity
- relaxer test for determining the hair’s reaction to the chemical and processing time

table 17-4
SELECTING THE CORRECT RELAXER

Active Ingredient	pH	Marketed as	Advantages	Disadvantages
Sodium hydroxide	12.5 to 13.5	Lye relaxer	Very effective for extremely curly hair	May cause scalp irritation and damage the hair
Lithium hydroxide and potassium hydroxide	12.5 to 13.5	No-mix, lye relaxer	Very effective for extremely curly hair	May cause scalp irritation and damage the hair
Guanidine hydroxide	13 to 13.5	No-lye relaxer	Causes less skin irritation than other hydroxide relaxers	More drying to hair with repeated use
Ammonium thioglycolate	9.6 to 10.0	Thio relaxer, no-lye relaxer	Compatible with soft-curl permanent waves	Strong, unpleasant ammonia smell
Ammonium sulfite or ammonium bisulfite	6.5 to 8.5	Low-pH relaxer, no-lye relaxer	Less damaging to hair	Does not relax extremely curly hair sufficiently

Relaxer Strand Test Mini-Procedure

1. Thread a small section of hair through a hole cut into a piece of wax paper or paper towel; do not use foil. Do not use a base or cream. Apply relaxer to the hair section and smooth. Process according to the manufacturer's directions.
 2. Thoroughly mist the hair section with water to remove product and blot.
 3. Neutralize with neutralizer or neutralizing shampoo depending on the type of relaxing product used. Check results and note on the client record card.
 4. If the hair has been satisfactorily straightened, apply conditioner or base cream to the strand, isolate it, and proceed with the relaxing treatment over the remainder of the hair.
-

SAFETY PRECAUTIONS FOR CHEMICAL RELAXERS

CAUTION

Hydroxide relaxers are incompatible with thio relaxers and products. *Never* use a hydroxide product on hair that has been previously relaxed with a thio product. *Never* use a thio product on hair that has been previously relaxed with a hydroxide product. Crossover of the products will result in hair breakage.

- Always protect the client's clothing with a proper waterproof cape.
- Use two towels, one under the drape and one over the cape.
- Always examine the client's scalp before a relaxer service. Do not proceed if abrasions are present.
- Do not proceed if the client has ever experienced an allergic reaction to the products.
- Do not relax excessively damaged hair.
- Do not use thio relaxers on hair that has been treated with hydroxide relaxers, or hydroxide relaxers on thio-treated hair.
- Always apply a protective cream barrier around the client's hairline before applying a relaxer.
- Base the scalp with a protective cream as directed by the product manufacturer.
- Always be careful of the client's eyes when applying relaxers and neutralizing solutions. In case of accidental exposure, rinse thoroughly with cool water.
- Always follow the manufacturer's directions.
- Do not dilute or add anything to relaxer creams unless specified in the manufacturer's directions.
- Wear gloves when applying relaxers.
- Do not save mixed products, as the strength and effectiveness will change if not used promptly.
- Apply relaxer cream to the most resistant area first.
- Follow the same pattern for smoothing the relaxer as was used during the application process.
- Frequently change towels as they become soiled or saturated.

Know about Chemical Curl Reformations

After reading this section, you will be able to:

LO 11 Explain the three steps of a curl reformation process.

WHAT IS A CURL REFORMATION?

A curl reformation, also known as a *soft-curl permanent*, is a three-step process that is used to restructure very curly hair into looser and larger curls. First, naturally curly hair is relaxed into a straighter form using a thio-based rearranger. This step is similar to a thio-based relaxer application. After processing and rinsing, the hair is wrapped on perm rods using a thio-based curl booster, much like a lotion wrap permanent wave wrapping procedure. Finally, hair is neutralized, rehardening the bonds in the cortex into a new, wavy pattern. A curl reformation is a combination of a thio-based relaxer application followed by a permanent wave wrap and neutralizing process.

The products used in a curl reformation include:

- *Rearranger*
 - The thio relaxer product used for the partial relaxation of the hair in a curl reformation service is most commonly known as the *rearranger*.
 - The rearranger is an ATG product in a thick cream form that is applied to dry hair in the same way as a hydroxide relaxer.
- *Booster*
 - The waving solution is called the *booster* and is similar in composition to alkaline permanent waving solutions or lotions.
 - This chemical restructures the hair around perm rods.
- *Neutralizer*
 - The *neutralizer* is an oxidizing solution in a clear or milky liquid form and is part of the same product line as the rearranger and the booster.
 - This solution rebuilds the broken disulfide bonds.
 - The chemical reactions of the neutralizer are the same as those created by permanent waving neutralizers.
- *Additional products*
 - Styling products such as activators and moisturizers are often recommended as part of the finishing process.
 - These products, such as activators and moisturizers, serve as leave-in conditioners and styling products to promote curl retention, although excess application should be avoided to prevent drips and oversaturation.
 - Depending on the texture and condition of the hair, other professional products that retain moisture and promote manageability of the hair may be preferred.



CAUTION

Use only the manufacturer's recommended neutralizer with the waving lotion. Never mix brands or systems.

Understand Texturizers and Chemical Blowouts

After reading this section, you will be able to:

LO12

Describe the intended outcomes of texturizer and chemical blowout services.

WHAT ARE THEY?

Overly curly hair has unique characteristics that may require special styling techniques. Relaxing products can be used to **texturize** the hair, to reduce it from an extremely curly state to a wave formation that makes haircutting and styling more versatile, or to perform a **chemical blowout** service that partially straightens the hair with the intent that it will be picked out and cut. Chemical blowouts were used for many of the semi-straightened “Afro” styles of the 1970s.



CAUTION

Review the neutralization procedures for thio and hydroxide relaxers at the end of each procedure before proceeding.

Using Thio

Texturizers and chemical blowouts are usually performed with a thio relaxer, although they may be performed with a sodium hydroxide relaxer. The primary consideration with either method is to not over-relax the hair to the point where the blowout process becomes impossible to perform. Consider the characteristics associated with using thio and hydroxide relaxers: Although thio relaxers may have a pH above 10, they are not as harsh as hydroxide relaxers. Depending on the strength of the thio relaxer, it may not straighten tightly curled hair textures to the desired degree. Since the objective of a chemical blowout is to remove some but not all of the curl, a thio relaxer may be used for most of its entire recommended application time. Check the curl pattern every few minutes until the desired amount of curl relaxation has been achieved.

P 17-3 Applying Thio Relaxer to Virgin Hair *pages 619–622*

P 17-4 Thio Relaxer Retouch *pages 623–625*

Using Hydroxide

Sodium hydroxide is an extremely alkaline product and may process the hair too quickly to the point where a blowout cannot be performed. When using sodium hydroxide, timing and controlling the amount of processing becomes very important. The chemical should not be kept on the hair for longer than 40 percent of the recommended processing time.

P 17-5 Applying Hydroxide Relaxer to Virgin Hair *pages 626–628*

P 17-6 Hydroxide Relaxer Retouch *pages 629–631*

Reminders

The important consideration to remember is to relax the hair only to the point where it has relaxed enough to achieve the desired style.

After the relaxer has been rinsed from the hair, apply:

- neutralizer conditioner if using a thio relaxer, or
- neutralizing (or stabilizing) shampoo and conditioner if using a sodium hydroxide product.

In both cases, the conditioner will help minimize possible damage or breakage and enable the hair to withstand combing, picking, and rearranging.

Procedural Reminders for a Chemical Blowout

While the degree of curl remaining after a texturizer service should leave the hair ready for cutting and styling, the chemical blowout requires an extra step before a haircut. After processing, perform the final steps of a chemical blowout service by:

1. Using a wide-tooth comb or pick, comb the hair upward and slightly forward. The hair closest to the scalp gives direction to the hair; therefore, it must be picked upward and outward. Start at the crown and continue until all of the hair has been combed out from the scalp and distributed evenly around the head.
2. Place the client under a hood hair dryer until the hair is dried.
3. Once dried, the hair is ready for shaping. Evenness is very important at this point. Check the hair length to make sure that the shortest hair is used as the guide for the balance of the head.
4. Begin cutting at the sides. The hair is evened out around the head with clippers or shears while picking the hair outward from the scalp. Cut the hair in the direction in which it is to be combed. The objective is to achieve a smooth, even cut that is properly contoured. The final cutting should be done with shears to even out loose or ragged ends.
5. Outline the hairstyle at the sides, around the ears, and in the nape area using either shears or an outliner. After the hair is cut to the desired style, apply the finishing touches. Fluff the hair slightly with the pick where required and spray lightly to hold the shape.



CAUTION

You may hear of keratin-based straightening treatments, also known as Brazilian keratin treatments. These advanced services are used to eliminate up to 95 percent of frizz and curl and last three to five months. If your state allows this service, educate yourself in order to protect you and your clients by following the requirements in Occupational Safety and Health Administration's (OSHA) formaldehyde and hazard communication standards. Potential health risks are involved if the formaldehyde released during the process exceeds the maximum concentration of 0.75 parts per million (ppm) allowed by OSHA over an 8-hour period. For more information, visit OSHA's website at <http://www.osha.gov>.

PRELIMINARY TEST CURL FOR A PERMANENT WAVE

MATERIALS, IMPLEMENTS, AND EQUIPMENT

You will need all of the following implements, materials, and supplies:

- ☐ Applicator bottles
- ☐ Clarifying and acid-balanced shampoo (optional)
- ☐ Conditioner (optional)
- ☐ Cotton coil or rope
- ☐ Disposable gloves
- ☐ End papers
- ☐ Neutralizer
- ☐ Neutralizing bib
- ☐ Perm rods
- ☐ Perm solution
- ☐ Plastic clips for sectioning
- ☐ Plastic tail comb
- ☐ Pre-neutralizing conditioner (optional)
- ☐ Protective barrier cream
- ☐ Roller picks
- ☐ Spray bottle
- ☐ Styling comb
- ☐ Timer
- ☐ Towels
- ☐ Waterproof cape

PREPARATION

- 1 Wash your hands.
- 2 Clean and disinfect all tools and implements.
- 3 Conduct client consultation.
- 4 Perform a hair and scalp analysis.
- 5 Select and arrange required materials.

PROCEDURE

- 1 Drape the client for shampoo.
- 2 Gently shampoo and towel-dry hair. Avoid irritating the client's scalp.
Re-drape the client for a chemical service.



- 3 Wrap one rod in each different area of the head (top, side, and nape).
- 4 Wrap a coil of cotton around each rod.



- 5 Apply perm solution to the wrapped curls. Do not allow perm solution to come into contact with unwrapped hair.
- 6 Set a timer, and process according to the manufacturer's directions.
- 7 Check each test curl frequently for proper curl development. Unfasten the rod and unwind the curl about one to two turns of the rod. Do not allow the hair to become loose or completely unwound. Gently move the rod toward the scalp to encourage the hair to fall loosely into the wave pattern.



- 8 Curl development is complete when a firm S is formed that reflects the size of the rod used. Different hair textures will have slightly different S formations. The wave pattern for fine, thin hair may be weak, with little definition. The wave pattern for coarse, thick hair is usually stronger and better defined.

- 9 When the desired curl has been formed, rinse thoroughly with warm water for at least 5 minutes, blot thoroughly, apply neutralizer, and process according to the manufacturer's directions. Gently dry the hair and evaluate the results. Do not proceed with the permanent wave if the test curls are extremely damaged or overprocessed. If the test curl results are satisfactory, proceed with the perm, but do not re-perm these preliminary test curls. Rinse and process the test rods, but wait to remove them with the rest of the rods after the perm is completed.

CLEAN-UP AND DISINFECTION

- ☐ Clean and disinfect tools and implements.
- ☐ Clean and disinfect work area.
- ☐ Dispose of single-use items. Place all used linens, towels, and capes in the laundry.
- ☐ Wash your hands.

CHEMICAL (PERMANENT) WAVE AND PROCESSING USING A BASIC PERMANENT WRAP

MATERIALS, IMPLEMENTS, AND EQUIPMENT

You will need all of the following implements, materials, and supplies:

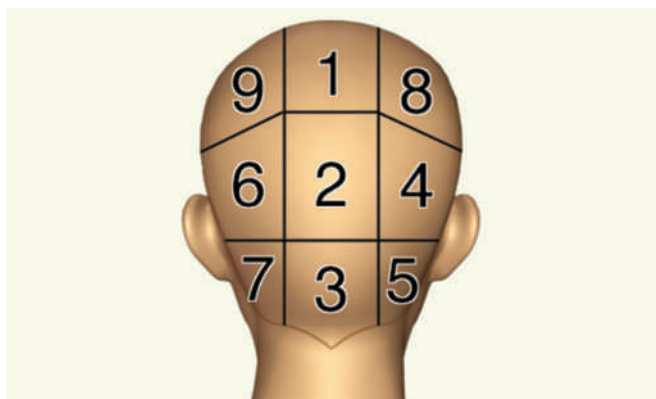
- ☐ Applicator bottles
- ☐ Clarifying and acid-balanced shampoo (optional)
- ☐ Conditioner (optional)
- ☐ Cotton coil or rope
- ☐ Disposable gloves
- ☐ End papers
- ☐ Neutralizer
- ☐ Neutralizing bib
- ☐ Perm rods
- ☐ Perm solution
- ☐ Plastic clips for sectioning
- ☐ Plastic tail comb
- ☐ Pre-neutralizing conditioner (optional)
- ☐ Protective barrier cream
- ☐ Roller picks
- ☐ Spray bottle
- ☐ Styling comb
- ☐ Timer
- ☐ Towels
- ☐ Waterproof cape

PREPARATION

- 1 Wash your hands.
- 2 Clean and disinfect all tools and implements.
- 3 Conduct client consultation.
- 4 Perform a hair and scalp analysis.
- 5 Select and arrange required materials.

PROCEDURE

- 1 After completing the pre-service procedure, seat the client. If the manufacturer's directions indicate a shampoo is necessary before the service, then drape the client for a shampoo and gently shampoo and towel-dry hair. Avoid irritating the client's scalp.
- 2 Re-drape the client for a chemical service.



- 3 Divide the hair into nine panels. Use the length of the rod to measure the width of the panels. Remember to keep the hair evenly damp as you wrap.



4A Begin wrapping at the front hairline or crown. Make a horizontal parting the same size as the rod. Using two end papers, roll the hair down to the scalp in the direction of hair growth, and position the rod half off-base.



4B The band should be smooth, not twisted, and should be fastened straight across the top of the rod. Excessive tension may cause band marks or hair breakage.



4C Continue wrapping the remainder of the first panel using the same technique. Option: Insert roller picks to stabilize the rods and eliminate any tension caused by the band.



5 Continue wrapping the remaining eight panels in numerical order, holding the hair at a 90-degree angle.



6 Apply protective barrier cream to the hairline and the ears. Apply a coil of cotton around the entire hairline and offer the client a towel to blot any drips. Put on gloves.



7 Slowly and carefully apply the perm solution to each rod. Ask the client to lean forward while you apply solution to the back area; ask the client to lean back as you apply solution to the front and sides. Avoid splashing and dripping. Continue to apply the solution slowly until each rod is completely saturated. Apply solution to the most resistant area first.



⑧ If a plastic cap is used, punch a few holes in the cap and cover all the hair completely. Do not allow the plastic cap to touch the client's skin.

⑨ Check cotton and towels. If they are saturated with solution, replace them.

⑩ Process according to the manufacturer's directions. Processing time varies according to the strength of the solution, hair type and condition, and desired results. As a general rule, processing usually takes less than 20 minutes at room temperature.



⑪ Check frequently for curl development. Unwind the rod and check the S pattern formation described in the preliminary test curl procedure. Check a different rod each time!



⑫ When processing is complete, rinse the hair thoroughly for at least 5 minutes. Then towel-blot each rod to remove excess moisture. Option: Some manufacturers recommend the application of a pre-neutralizing conditioner after rinsing and blotting and before applying the neutralizer. Always follow the manufacturer's directions and the procedures approved by your instructor.



13 Apply the neutralizer slowly and carefully to the hair on each rod. Ask the client to lean forward while you apply solution to the back area, and then to lean back as you apply solution to the front and sides. Avoid splashing and dripping. Continue to apply the neutralizer until each rod is completely saturated.

14 Set a timer for the amount of time specified by the manufacturer.

15 Rinse thoroughly. Option: Shampoo and condition. Always follow the manufacturer's directions and the procedures approved by your instructor.



16 Style the hair as desired.

CLEAN-UP AND DISINFECTION

- ☐ Clean and disinfect tools and implements.
- ☐ Clean and disinfect work area.
- ☐ Dispose of single-use items. Place all used linens, towels, and capes in the laundry.
- ☐ Wash your hands.

APPLYING THIO RELAXER TO VIRGIN HAIR

MATERIALS, IMPLEMENTS, AND EQUIPMENT

You will need all of the following implements, materials, and supplies:

- | | |
|--|--|
| ☐ Acid-balanced shampoo | ☐ Protective base cream |
| ☐ Applicator brush or tail comb | ☐ Spray bottle |
| ☐ Conditioner | ☐ Styling comb |
| ☐ Disposable gloves | ☐ Thio neutralizer |
| ☐ Hard rubber comb | ☐ Thio relaxer |
| ☐ Plastic clips | ☐ Timer |
| ☐ Plastic or glass bowl | ☐ Towels |
| ☐ Pre-neutralizing conditioner | ☐ Waterproof cape |

PREPARATION

- 1 Wash your hands.
- 2 Clean and disinfect all tools and implements.
- 3 Conduct client consultation.

PROCEDURE

- 1 Perform an analysis of the hair and scalp. Perform tests for porosity and elasticity.



- 2 Drape the client for a chemical service. *The hair and scalp must be completely dry prior to the application of a thio relaxer.*



- 3 Part the hair into four sections, from the center of the front hairline to the center of the nape, and from ear to ear. Clip the sections up to keep them out of the way.





4 Apply protective base cream to the hairline and ears. Option: Take ¼-inch to ½-inch horizontal partings, and apply a protective base cream to the entire scalp. Always follow the manufacturer's directions and the procedures approved by your instructor.



5 Wear gloves on both hands. Begin application in the most resistant area, usually at the back of the head. Make ¼-inch to ½-inch horizontal partings, and apply the relaxer to the top of the strand first, and then to the underside. Apply the relaxer with an applicator brush, with the back of the comb, or with your fingers. Apply relaxer ¼-inch to ½-inch away from the scalp, and up to the porous ends. To avoid scalp irritation, do not allow the relaxer to touch the scalp until the last few minutes of processing.



6 Continue applying the relaxer, working your way down the section toward the hairline.



7 Continue the same application procedure with the remaining sections. Finish the most resistant sections first.



8 After the relaxer has been applied to all sections, use the back of the comb or your hands to smooth each section. Never comb the relaxer through the hair.

9 Process according to the manufacturer's directions. Perform periodic strand tests. Processing usually takes less than 20 minutes at room temperature. Always follow the manufacturer's processing directions.



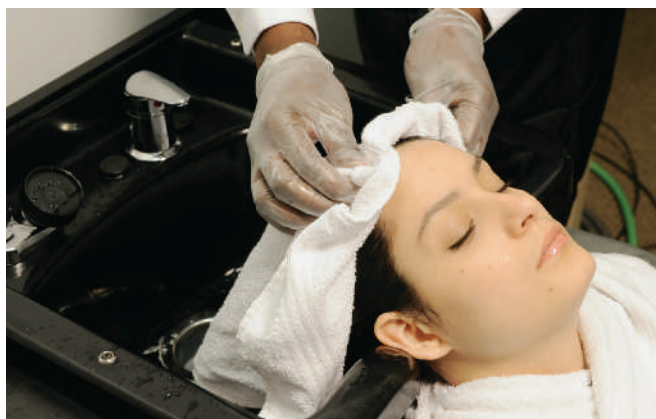
10 During the last few minutes of processing, work the relaxer down to the scalp and through the ends of the hair, using additional relaxer as needed. Carefully smooth all sections using an applicator brush, your fingers, or the back of the comb.



11 Rinse thoroughly with warm water to remove all traces of the relaxer.



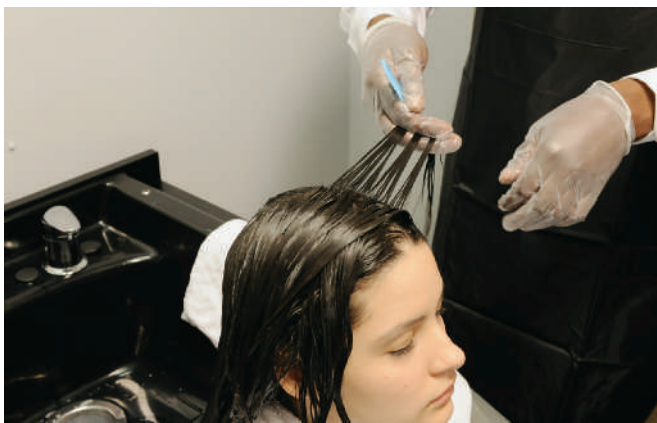
12 Shampoo at least three times with an acid-balanced shampoo. It is essential that all traces of the relaxer be removed from the hair. Optional: Apply the pre-neutralizing conditioner, and comb it through to the ends of the hair. Leave it on for approximately 5 minutes and then rinse. Always follow the manufacturer's directions and the procedures approved by your instructor.



13 Blot excess water from the hair.



14 Apply thio neutralizer in ¼-inch to ½-inch sections throughout the hair and smooth with your hands or the back of the comb.



15 Process the neutralizer according to the manufacturer's directions.



16 Rinse thoroughly. Shampoo, condition, and style.

CLEAN-UP AND DISINFECTION

- ☐ Clean and disinfect tools and implements.
- ☐ Clean and disinfect work area.
- ☐ Dispose of single-use items. Place all used linens, towels, and capes in the laundry.
- ☐ Wash your hands.

THIO RELAXER RETOUCH

MATERIALS, IMPLEMENTS, AND EQUIPMENT

You will need all of the following implements, materials, and supplies:

- ☐ Acid-balanced shampoo
- ☐ Applicator brush or tail comb
- ☐ Conditioner
- ☐ Disposable gloves
- ☐ Hard rubber comb
- ☐ Plastic clips
- ☐ Plastic or glass bowl
- ☐ Pre-neutralizing conditioner
- ☐ Protective base cream
- ☐ Spray bottle
- ☐ Styling comb
- ☐ Thio neutralizer
- ☐ Thio relaxer
- ☐ Timer
- ☐ Towels
- ☐ Waterproof cape

Photography by Tom Carson.



PREPARATION

- 1 Wash your hands.
- 2 Clean and disinfect all tools and implements.
- 3 Conduct client consultation.
- 4 Perform a hair and scalp analysis.

PROCEDURE

- 1 Perform an analysis of the hair and scalp. Perform tests for porosity and elasticity.
- 2 Drape the client for a chemical service. To avoid scalp irritation, do not shampoo the hair prior to a thio relaxer. *The hair and scalp must be completely dry prior to the application of a thio relaxer retouch.*



- 3 Divide the hair into four sections, from the center of the front hairline to the center of the nape, and from ear to ear. Clip sections up to keep them out of the way.



4 Wear gloves on both hands. Apply a protective base cream to the hairline and ears, unless you are using a no-base relaxing product. Option: Take ¼-inch to ½-inch horizontal partings, and apply protective base cream to the entire scalp.



5 Begin application of the relaxer in the most resistant area, usually at the back of the head. Make ¼-inch to ½-inch horizontal partings, and apply the relaxer to the top of the strand. Apply the relaxer as close to the scalp as possible, but do not touch the scalp with the product. Only allow the relaxer to touch the scalp during the last few minutes of processing. To avoid overprocessing or breakage, do not overlap the relaxer onto the previously relaxed hair.



6 Continue applying the relaxer, using the same procedure and working your way down the section toward the hairline.



7 Continue the same application procedure with the remaining sections, finishing the most resistant sections first.



8 After the relaxer has been applied to all sections, use the back of the comb, the applicator brush, or your hands to smooth each section.

9 Process according to the manufacturer's directions. Perform periodic strand tests. Processing usually takes less than 20 minutes at room temperature. Always follow the manufacturer's processing directions.



- ⑩ During the last few minutes of processing, gently work the relaxer down to the scalp.

**CAUTION**

Never intentionally overlap previously relaxed hair as this will result in damage and possible hair breakage!

- ⑪ Rinse thoroughly with warm water to remove all traces of the relaxer.
- ⑫ Shampoo at least three times with an acid-balanced shampoo (neutralizer). It is essential that all traces of the relaxer be removed from the hair. If the relaxer product recommends using a pre-neutralizing conditioner, comb it through the hair as per the recommendations of the manufacturer. Always follow the manufacturer's directions and the procedures approved by your instructor.
- ⑬ Blot excess water from hair.



- ⑭ Apply thio neutralizer in ¼-inch to ½-inch sections throughout the hair and smooth with your hands or the back of the comb.
- ⑮ Process the neutralizer according to the manufacturer's directions.
- ⑯ Rinse thoroughly. Shampoo, condition, and style.

CLEAN-UP AND DISINFECTION

- ☐ Clean and disinfect tools and implements.
- ☐ Clean and disinfect work area.
- ☐ Dispose of single-use items. Place all used linens, towels, and capes in the laundry.
- ☐ Wash your hands.

APPLYING HYDROXIDE RELAXER TO VIRGIN HAIR



MATERIALS, IMPLEMENTS, AND EQUIPMENT

You will need all of the following implements, materials, and supplies:

- ☐ Conditioner
- ☐ Disposable gloves
- ☐ Hydroxide relaxer
- ☐ Neutralizing acid-balanced shampoo
- ☐ Plastic clips
- ☐ Plastic or glass bowl
- ☐ Protective base cream
- ☐ Styling comb
- ☐ Tail comb or applicator brush
- ☐ Timer
- ☐ Towels
- ☐ Waterproof cape
- ☐ Wide-tooth hard rubber comb

PREPARATION

- 1 Wash your hands.
- 2 Clean and disinfect all tools and implements.
- 3 Conduct client consultation.
- 4 Perform a hair and scalp analysis.
- 5 Select and arrange required materials.
- 6 Drape client for a chemical service.

PROCEDURE

- 1 Perform an analysis of the hair and scalp by visually assessing the hair for breakage, sores on the scalp, or any visual signs of irritation. Feel the hair and perform an elasticity test. If hair fails the test for porosity and elasticity, do not perform the relaxer service.



- 2 Drape the client for a chemical service.



- 3 To avoid scalp irritation, do not shampoo the hair. *The hair and scalp must be completely dry prior to the application of a hydroxide relaxer.*



4A Part the hair into four sections, from the center of the front hairline to the center of the nape, and from ear to ear. Clip the sections up if necessary to keep hair out of the way.



4B Apply protective base cream to the hairline and ears.

4C Optional: Take ¼-inch to ½-inch horizontal partings, and apply a protective base cream to the entire scalp. Always follow the manufacturer's directions and the procedures approved by your instructor. Set timer as indicated by the manufacturer and initial strand test.



5A Put gloves on both hands. Begin the relaxer application in the most resistant area, usually at the back of the head or nape area. Make ¼-inch to ½-inch horizontal partings, and apply the relaxer to the top of the strand first. Do not apply to the scalp.



5B Apply relaxer to the underside of the first section using an applicator brush or the back of a tail comb. Apply relaxer ¼-inch to ½-inch away from the scalp, and up to the porous ends. To avoid scalp irritation, do not allow the relaxer to touch the scalp until the last few minutes of processing.

6 Continue applying relaxer to other sections, working your way down the section toward the hairline. Continue the same application procedure with the remaining sections.



7 After the relaxer has been applied to all sections, use the back of the comb or your hands to smooth each section. Never comb the relaxer through the hair as this may break the hair.



8 Process according to the manufacturer's directions and what your initial strand test indicated regarding timing. Perform periodic strand tests.



9 During the last few minutes of processing, work the relaxer down to the scalp and through the ends of the hair, using additional relaxer as needed. Carefully smooth all sections, using an applicator brush, your fingers, or the back of the tail comb.

10 Rinse thoroughly with warm water to remove all traces of the relaxer.

11 If the relaxer comes with a normalizing lotion or conditioner, comb it through the hair. Leave it on as indicated by the manufacturer. Rinse thoroughly. Always follow the manufacturer's directions and the procedures approved by your instructor.

12 Shampoo at least three times with an acid-balanced neutralizing shampoo. If you are using a neutralizing shampoo with a color indicator, usually the color will change from pink to white indicating that all traces of the relaxer are removed, and the natural pH of the hair and scalp has been restored.



13 Rinse thoroughly, condition, and style as desired.

CLEAN-UP AND DISINFECTION

- ☐ Clean and disinfect tools and implements.
- ☐ Clean and disinfect work area.
- ☐ Dispose of any single use items, paper goods, and/or linens.
- ☐ Wash your hands.

HYDROXIDE RELAXER RETOUCH

MATERIALS, IMPLEMENTS, AND EQUIPMENT

You will need all of the following implements, materials, and supplies:

- ☐ Acid-balanced shampoo
- ☐ Conditioner
- ☐ Disposable gloves
- ☐ Hard rubber comb
- ☐ Hydroxide neutralizer
- ☐ Hydroxide relaxer
- ☐ Plastic clips
- ☐ Plastic or glass bowl
- ☐ Protective base cream
- ☐ Spray bottle
- ☐ Tail comb or applicator brush
- ☐ Timer
- ☐ Towels
- ☐ Waterproof cape



PREPARATION

- 1 Wash your hands.
- 2 Clean and disinfect all tools and implements.
- 3 Conduct client consultation.

PROCEDURE

- 1 Perform an analysis of the hair and scalp. Perform tests for porosity and elasticity.
- 2 Drape the client for a chemical service.
- 3 To avoid scalp irritation, do not shampoo the hair. *The hair and scalp must be completely dry prior to the application of a hydroxide relaxer retouch.*



- 4A Divide the hair into four sections, from the center of the front hairline to the center of the nape, and from ear to ear. Clip sections up to keep hair out of the way if necessary.



- 4B Apply a protective base cream to the hairline and ears. Put gloves on both hands.
- 4C Optional: Take ¼-inch to ½-inch horizontal partings and apply protective base cream to the entire scalp. Set timer as indicated by the manufacturer for retouch.



⑤ Begin application of the relaxer in the most resistant area, usually at the back of the head or nape area. Make ¼-inch to ½-inch horizontal partings, and apply the relaxer to the top of the strand. Apply the relaxer as close to the scalp as possible, but do not touch the scalp with the relaxer. Only allow the relaxer to touch the scalp during the last few minutes of processing. To avoid overprocessing or breakage, do not overlap the relaxer onto the previously relaxed hair.

⑥ Continue applying the relaxer, using the same procedure, and working your way down the section toward the hairline.



⑦ Continue the same application procedure with the remaining sections.



⑧ After the relaxer has been applied to all sections, use the back of the comb, the applicator brush, or your hands to smooth each section.

⑨ Process according to the manufacturer's directions. Perform periodic strand tests.

⑩ During the last few minutes of processing, gently work the relaxer down to the scalp.